FORMAL COMPLETION OF AN AFFINE FORMAL ALGEBRAIC SPACE ALONG A CLOSED SUBSET KEEPS ITS TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Part (4) of the proof ends with the author's 'Some details omitted.' It is reproduced verbatim and recorded as an obligation of the card; the parallel argument in part (3) is carried out in full and part (4) is explicitly said to be the same argument in the adic case.

Editorial note. The proof displays three descriptions of the completion as a limit of quotients; all lie inside the counted span and are reproduced verbatim.

Editorial note. The counted spans carry 6 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The formal completion of an affine formal algebraic space along a closed subset is affine. Same chapter, Lemma lemma-completion-affine-formal-is-affine-formal, invoked at proof lines 8122-8123. Gives part (1).
- The second defining property of a McQuillan affine formal algebraic space. Same chapter, Lemma lemma-mcquillan-affine-formal-algebraic-space at proof lines 8124-8127 and Definition definition-types-affine-formal-algebraic-space at proof lines 8127-8128. Gives part (2).
- Countably indexed affine formal algebraic spaces as formal spectra with a countable fundamental system of weak ideals of definition. Same chapter, Lemma lemma-countably-indexed, invoked at proof lines 8131-8134. Supplies the presentation used in the computation for part (3).
- A closed subset with quasi-compact complement in an affine scheme is cut out by a finitely generated ideal. Algebra, Lemma algebra-lemma-qc-open, invoked at proof lines 8134-8137. Produces the finitely generated ideal whose lifts are used to describe the completion.
- Completions of Noetherian rings. Algebra, Lemma algebra-lemma-completion-Noetherian-Noetherian, invoked at proof lines 8164-8165. Gives part (5) once the completion has been described as a limit of quotients.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0316: sources/stacks-chow/tags.txt, source lines 3973–3973. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00F6: sources/stacks-chow/tags.txt, source lines 560–560. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GVQ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7969–7969. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN1: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2493–2493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIC: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2135–2135. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 07E8: sources/stacks-chow/tags.txt, source lines 9438–9438. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 0GXE: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1595–1595. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. Let X be an affine formal algebraic space over S. We say X is *McQuillan* if X satisfies the equivalent conditions of Lemma 0AIC. Let A be the weakly admissible topological ring associated to X. We say

(1) X is *classical* if X is McQuillan and A is admissible (More on Algebra, Definition 07E8),
(2) X is *weakly adic* if X is McQuillan and A is weakly adic (Definition 0GXE),
(3) X is *adic* if X is McQuillan and A is adic (More on Algebra, Definition 07E8),
(4) X is *adic** if X is McQuillan, A is adic, and A has a finitely generated ideal of definition, and
(5) X is *Noetherian* if X is McQuillan and A is both Noetherian and adic.

Lemma 0.2. *Let S be a scheme. Let X be an affine formal algebraic space over S. Let $T \subset X_{red}$ be a closed subset and let $X_{/T}$ be the formal completion of X along T. Then*

(1) *$X_{/T}$ is an affine formal algebraic space,*
(2) *if X is McQuillan, then $X_{/T}$ is McQuillan,*

(3) *if $|X_{red}| \setminus T$ is quasi-compact and X is countably indexed, then $X_{/T}$ is countably indexed,*
(4) *if $|X_{red}| \setminus T$ is quasi-compact and X is adic*, then $X_{/T}$ is adic*,*
(5) *if X is Noetherian, then $X_{/T}$ is Noetherian.*

Proof. Part (1) is Lemma 0GVQ. If X is McQuillan, then $X = \mathrm{Spf}(A)$ for some weakly admissible topological ring A. Then $X_{/T} \to X \to \mathrm{Spec}(A)$ satisfies property (2) of Lemma 0AIC and hence $X_{/T}$ is McQuillan, see Definition 0.1.

Assume X and T are as in (3). Then $X = \mathrm{Spf}(A)$ where A has a fundamental system $A \supset I_1 \supset I_2 \supset I_3 \supset \ldots$ of weak ideals of definition, see Lemma 0AN1. By Algebra, Lemma 00F6 we can find a finitely generated ideal $\overline{J} = (\overline{f}_1, \ldots, \overline{f}_r) \subset A/I_1$ such that T is cut out by $\overline{J}$ inside $\mathrm{Spec}(A/I_1) = |X_{red}|$. Choose $f_i \in A$ lifting $\overline{f}_i$. If $Z = \mathrm{Spec}(B)$ is an affine scheme and $g : Z \to X$ is a morphism with $g(Z) \subset T$ (set theoretically), then $g^\sharp : A \to B$ factors through A/I_n for some n and $g^\sharp(f_i)$ is nilpotent in B for each i. Thus $J_{m,n} = (f_1, \ldots, f_r)^m + I_n$ maps to zero in B for some $n, m \geq 1$. It follows that $X_{/T}$ is the formal spectrum of $\lim_{n,m} A/J_{m,n}$ and hence countably indexed. This proves (3).

Proof of (4). Here the argument is the same as in (3). However, here we may choose $I_n = I^n$ for some finitely generated ideal $I \subset A$. Then it is clear that $X_{/T}$ is the formal spectrum of $\lim A/J^n$ where $J = (f_1, \ldots, f_r) + I$. Some details omitted.

Proof of (5). In this case X_{red} is the spectrum of a Noetherian ring and hence the assumption that $|X_{red}| \setminus T$ is quasi-compact is satisfied. Thus as in the proof of (4) we see that $X_{/T}$ is the spectrum of $\lim A/J^n$ which is a Noetherian adic topological ring, see Algebra, Lemma 0316. □